

Introduction to Pandas

Data Analysis with Python

What is Pandas? 🐼

Pandas is Python's most popular library for data analysis and manipulation

- Built on top of NumPy
- Provides easy-to-use data structures
- Essential for data science workflows
- Used by millions of data professionals

"Pandas makes working with data feel natural and intuitive"

Why Learn Pandas?

- **Industry Standard** - Used in data science, finance, research, and more
- **Powerful** - Handle datasets with millions of rows
- **Versatile** - Read/write Excel, CSV, SQL, JSON, and more
- **Fast** - Optimized C code under the hood
- **Open Source** - Free and constantly improving

Core Data Structures

DataFrame - The main workhorse

- 2D table with rows and columns
- Like an Excel spreadsheet, but programmable
- Each column can have different data types

Series - A single column

- 1D array with labels
- Think of it as one column from a DataFrame

Our Dataset: Irish Traditional Music 🎵

We'll be working with a real dataset of **24,000 traditional Irish tunes**

Source: The Session (thesession.org) - world's largest collection of Irish traditional music

Format: ABC notation - a text-based music notation system

Dataset Features

What's in our data?

Column	Description	Example
id	Unique identifier	1, 2, 3...
title	Tune name	"Cooley's"
tune_type	Category	reel, jig, hornpipe
key_sig	Musical key	Emin, Dmaj, Gmaj
time_sig	Time signature	4/4, 6/8, 9/8
downloaded	Popularity metric	4432, 2721...
notation	ABC music code	Full tune notation

Sample Data

id	title	tune_type	key_sig	downloads
1	Cooley's	reel	Emin	4432
2	Bucks Of Oranmore	reel	Dmaj	2721
3	Boil The Breakfast	reel	Gmaj	812
10	The Butterfly	slip jig	Emin	2767
12	Cliffs Of Moher	jig	Ador	2850

Types of Tunes in Irish Music

Reels (4/4 time)

Fast dance tunes, most common type

Jigs (6/8 time)

Bouncy, lilting rhythm

Hornpipes (4/4 time)

Slower than reels, swung rhythm

Slip Jigs (9/8 time)

Graceful, flowing tunes

Essential Pandas Operations

```
import pandas as pd

# Load data
df = pd.read_csv('tuneindex.csv')

# Explore
df.head()           # First 5 rows
df.info()           # Column types and info
df.describe()       # Statistical summary
```

Essential Pandas Operations (cont.)

```
# Select columns
df['title']           # Single column
df[['title', 'tune_type']] # Multiple columns

# Filter rows
df[df['tune_type'] == 'reel'] # Only reels
df[df['downloaded'] > 1000]   # Popular tunes

# Sort
df.sort_values('downloaded', ascending=False)
```

Essential Pandas Operations (cont.)

```
# Group and aggregate  
df.groupby('tune_type')['downloaded'].mean()
```

```
# Count values  
df['tune_type'].value_counts()
```

```
# Handle missing data  
df.dropna()           # Remove missing  
df.fillna(value)      # Fill missing
```

Questions We'll Answer

1. What's the most popular tune type?
2. Which key signature is most common?
3. Is there a relationship between key and popularity?
4. What makes a tune popular?
5. How do different tune types compare?
6. Are certain time signatures more popular?

Lesson 1: Load and Explore

```
import pandas as pd

# Your first pandas code!
df = pd.read_csv('tuneindex.csv')

print(f"We have {len(df)} tunes!")
print(f"Columns: {list(df.columns)}")
print(f"\nFirst tune:")
print(df.iloc[0])
```

Installing Pandas

```
pip install pandas
```

Verify Installation

```
import pandas as pd  
print(pd.__version__)
```

Let's Look at Real Data

```
import pandas as pd

df = pd.read_csv('tuneindex.csv')

# What's the most popular tune?
most_popular = df.nlargest(1, 'downloaded')
print(most_popular[['title', 'tune_type', 'downloaded']])

# Output:
#   title      tune_type  downloaded
#  9 Banish Misfortune  jig        4292
```

Common Pandas Patterns

You'll use these constantly:

```
# Load → Filter → Group → Visualize
df = pd.read_csv('data.csv')
result = df[df['tune_type'] == 'jig'].nlargest(1, 'downloaded')
```

Example: First Analysis

Question: How many tune types are in our dataset?

```
import pandas as pd

df = pd.read_csv('tuneindex.csv')
tune_types = df['tune_type'].value_counts()

print(tune_types)
```

Output:

```
reel      12450
jig        7832
hornpipe   2234
slip jig    892
...
```


The Power of Pandas

Before Pandas:

```
# Reading CSV manually - 50+ lines of code  
# Filtering data - loops and conditions  
# Grouping - complex dictionary logic  
# Plotting - connecting to other libraries
```

With Pandas:

```
df = pd.read_csv('data.csv')  
df[df['type'] == 'reel'].groupby('key')['downloads'].mean().plot()
```

One line does it all!

